MACHURA RAMSEY SHADOW THEORY - UNIFIED SIGNED VIEW
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Project root: C:\mcdo_ramsey_MASTER43\wzor_cienia_ramseya_machura

Scope:
This document collects the current theory layer of the Machura Ramsey Shadow Formula.
It uses standard Ramsey terminology and separates theory from numerical audit evidence.

Included source files:

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DOCUMENT BODY
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LEMAT ODWROCENIA CIENIA
=======================

Autor: Michal Machura

Niech C_n^{a,b} oznacza zbior czystych rdzeni Ramseyowskich.

Niech chi : V(G) -> {red, blue} oznacza sposob polaczenia nowego
wierzcholka z rdzeniem G.

Niech G + chi oznacza graf po dodaniu nowego wierzcholka.

Definiujemy:

Conf_{a,b}(G)
=
RedK_a(G) union BlueK_b(G^c)

oraz:

Phi_{a,b,G}(chi)

=
zbior nowych konfliktow powstalych z udzialem dodanego wierzcholka.

Wzor przyrostu:

Conf_{a,b}(G + chi)
=
Conf_{a,b}(G) union Phi_{a,b,G}(chi)

Dla czystego rdzenia:

Conf_{a,b}(G) = empty

zatem:

Conf_{a,b}(G + chi) = Phi_{a,b,G}(chi)

Lemat odwrocenia:

C_{succ(n)}^{a,b} != empty

wtedy i tylko wtedy, gdy istnieje G in C_n^{a,b}
oraz istnieje chi : V(G) -> {red, blue}
takie, ze:

Phi_{a,b,G}(chi) = empty.

Dowod w przod:

Jesli istnieje para (G, chi) z pustym cieniem, to:

Conf(G + chi)
=
Conf(G) union Phi_G(chi)
=
empty union empty
=
empty

czyli G + chi jest czysty.

Dowod w tyl:

Jesli istnieje czysty graf H na succ(n) wierzcholkach,
to po usunieciu dowolnego wierzcholka v dostajemy czysty rdzen:

G = H - v

Kolory krawedzi z v do G definiuja chi.

Poniewaz H byl czysty, dodany wierzcholek nie tworzy zadnego konfliktu,
wiec:

Phi_G(chi) = empty.

Wniosek:

Szukanie czystego grafu na succ(n) jest rownowazne szukaniu pustego cienia nad czystym rdzeniem na n.

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MACHURA RAMSEY SHADOW THEORY
V04 THEORETICAL CLOSURE WITHOUT NUMERICAL PARAMETERS
==

Author:
Michal Machura

Scope:
This document formulates the Machura Ramsey Shadow theory without using the numerical R(5,5) boundary values.

The purpose is to express the theory as a general Ramsey-theoretic one-vertex extension and deletion calculus.

No numerical Delta value is used in this document.
No enumeration result is used in this document.
No solver-specific mechanism is used in this document.

1. Standard Ramsey setting
==========================

Let R(a,b) denote the classical Ramsey number.

A graph G is called an (a,b)-clean graph if:

G contains no red K_a,

and the complement graph G^c contains no blue K_b.

Equivalently:

Conf_{a,b}(G) = empty,

where:

Conf_{a,b}(G)
=
RedK_a(G) union BlueK_b(G^c).

Let:

C_n^{a,b}
=
{G : |V(G)| = n and Conf_{a,b}(G) = empty}.

Thus C_n^{a,b} is the layer of clean Ramsey cores on n vertices.

2. One-vertex extension
=======================

Let G be a graph on vertex set V.

Let chi be a connection map from a new vertex to the old core:

chi : V(G) -> {red, blue}.

The extended graph is denoted:

G + chi.

The shadow Phi_{a,b,G}(chi) is defined as the set of all new forbidden Ramsey conflicts created with the participation of the added vertex.

Thus Phi contains exactly those forbidden structures which did not exist inside G alone and which become complete after adding the new vertex.

3. Forward shadow identity

For every graph G and every connection map chi:

Conf_{a,b}(G + chi)
=
Conf_{a,b}(G) union Phi_{a,b,G}(chi).

In score form:

score_{a,b}(G + chi)
=
score_{a,b}(G) + |Phi_{a,b,G}(chi)|.

If G is clean, then:

Conf_{a,b}(G) = empty,

so:

Conf_{a,b}(G + chi)
=
Phi_{a,b,G}(chi).

Therefore, for a clean core, the whole obstruction to a clean extension is exactly the shadow.

4. Reverse shadow identity

Let H be a graph and let v be a vertex of H.

Define:

G_v = H - v.

Let chi_v be the connection map induced by the deleted vertex v:

chi_v : V(G_v) -> {red, blue}.

Then:

$$\mathrm{Conf}_{a,b}(H) = \mathrm{Conf}_{a,b}(H - v) \cup \Phi_{a,b,H-v}(\chi_v).$$

In score form:

$$\mathrm{score}_{a,b}(H) = \mathrm{score}_{a,b}(H - v) + |\Phi_{a,b,H-v}(\chi_v)|.$$

Equivalently:

$$|\Phi_{a,b,H-v}(\chi_v)| = \mathrm{score}_{a,b}(H) - \mathrm{score}_{a,b}(H - v).$$

Interpretation:

The shadow of a deleted vertex is exactly the part of the Ramsey conflict score that disappears when the vertex is removed.

5. Vertex-sum identity

Summing the reverse shadow over all vertices of H gives:

$$\sum_{v \in V(H)} |\Phi_v(H)| = a \cdot |\mathrm{RedK}_a(H)| + b \cdot |\mathrm{BlueK}_b(H^c)|.$$

Reason:

Every red K_a has exactly a vertices, so it is counted once for each of its vertices.

Every blue K_b has exactly b vertices, so it is counted once for each of its vertices.

This gives a global consistency check for the shadow calculus.

6. Shadow reversal lemma

The following statements are equivalent:

A. There exists a clean graph H on succ(n) vertices.

B. There exists a clean core G in $C_n^{a,b}$ and a connection map chi such that:

$$\Phi_{a,b,G}(\chi) = \emptyset.$$

Proof from B to A:

If G is clean and $\Phi_{a,b,G}(\chi)$ is empty, then:

Conf(G + chi)

=

Conf(G) union Phi_G(chi)

=

empty union empty

=

empty.

Therefore G + chi is clean.

Proof from A to B:

If H is clean on succ(n) vertices, delete any vertex v.

Then:

G_v = H - v

is clean, because deleting a vertex cannot create a forbidden clique
or independent set.

The deleted vertex induces a connection map chi_v.

Since H is clean, the deleted vertex participates in no forbidden
Ramsey conflict.

Therefore:

Phi_{G_v}(chi_v) = empty.

Thus a clean larger graph exists if and only if a clean core with empty
shadow exists.

7. Shadow closure theorem

If:

C_n^{a,b} is not empty,

and:

for every G in C_n^{a,b}
and every connection map chi : V(G) -> {red, blue},

Phi_{a,b,G}(chi) is not empty,

then:

C_{succ(n)}^{a,b} is empty.

Therefore:

R(a,b) = succ(n).

This is the theoretical form of the Machura Shadow Closure.

8. Falsification criterion

==========================

To refute shadow closure at a given layer, it is enough to find one pair:

(G, chi)

such that:

G in C_n^{a,b},

and:

Phi_{a,b,G}(chi) = empty.

Such a pair immediately constructs a clean graph:

H = G + chi

on succ(n) vertices.

Thus the entire closure claim is falsified by one empty shadow.

9. Quantitative Delta layer

The qualitative theory requires only:

Phi_{a,b,G}(chi) is not empty.

The quantitative version defines:

Delta_{a,b}(n -> succ(n))
=
min_{G in C_n^{a,b}} min_chi |Phi_{a,b,G}(chi)|.

Then:

Delta_{a,b}(n -> succ(n)) > 0

is equivalent to shadow closure at that layer.

A concrete numerical value of Delta belongs to the evidence and audit layer, not to the abstract theory in this document.

10. Relation to validated tests

The forward and reverse identities were locally validated on real clean Ramsey cores by the following tests:

V01:
Shadow reversal validation.
Clean H gives clean deleted cores and empty reconstructed shadow.

V03:
Shadow difference validation.
The deleted-vertex shadow equals score(H) - score(H-v), including separate red and blue components and the vertex-sum identity.

These tests validate the internal consistency of the shadow calculus.
They do not replace a full layer enumeration.

11. Core sentence

A clean graph one vertex larger exists exactly when a clean core has an empty shadow.

Equivalently:

No empty shadow over the clean core layer means Ramsey closure at the next vertex.

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SHA256: d5389c99000c09ce1a1954d84dfadd8f977570ef8d6743441542603a6be1e032
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MACHURA RAMSEY SHADOW THEOREM MAP
V05 ONE VIEW

Author:
Michal Machura

Purpose:
One-view formal map of the Machura Ramsey Shadow theory using standard Ramsey-theoretic terminology.

This file is theoretical.
It does not contain a numerical proof.
It does not enumerate graphs.
It does not expose solver internals.

1. Standard Ramsey objects

Ramsey number:

R(a,b)

is the smallest integer N such that every graph on N vertices contains a clique K_a or an independent set of size b.

Equivalently, in two-color edge language:

every red-blue coloring of K_N contains a red K_a or a blue K_b.

Clean graph:

A graph G is called (a,b)-clean if:

G contains no clique K_a,

and:

G^c contains no clique K_b.

Equivalently:

G has no red K_a and no blue K_b.

Clean layer:

C_n^{a,b}
=
{G : |V(G)| = n and G is (a,b)-clean}.

2. Conflict set

Define the Ramsey conflict set:

Conf_{a,b}(G)
=
RedK_a(G) union BlueK_b(G^c).

A graph is clean if and only if:

Conf_{a,b}(G) = empty.

The score is the size of the conflict set:

score_{a,b}(G)
=
|Conf_{a,b}(G)|.

3. One-vertex extension

Let G be a clean core.

Let chi be a connection map from a new vertex to the old vertex set:

chi : V(G) -> {red, blue}.

The extended graph is:

G + chi.

4. Machura shadow

The shadow Phi_{a,b,G}(chi) is the set of all new forbidden Ramsey conflicts created by the added vertex.

It consists of:

red K_a created from red K_{a-1} structures in G,

and:

blue K_b created from blue K_{b-1} structures in G^c.

5. Forward identity
===================

For every graph G and every connection map chi:

Conf_{a,b}(G + chi)
=
Conf_{a,b}(G) union Phi_{a,b,G}(chi).

In score form:

score_{a,b}(G + chi)
=
score_{a,b}(G) + |Phi_{a,b,G}(chi)|.

For a clean core G:

score_{a,b}(G) = 0,

so:

score_{a,b}(G + chi)
=
|Phi_{a,b,G}(chi)|.

6. Reverse identity
===================

Let H be a graph and v in V(H).

Let:

G_v = H - v.

Let chi_v be the connection map induced by v to G_v.

Then:

score_{a,b}(H)
=
score_{a,b}(H - v)
+
|Phi_{a,b,H-v}(chi_v)|.

Equivalently:

|Phi_{a,b,H-v}(chi_v)|
=
score_{a,b}(H) - score_{a,b}(H - v).

Interpretation:

The shadow of v is exactly the part of the Ramsey conflict score
that disappears when v is removed.

7. Vertex-sum identity
======================

For every graph H:

sum_{v in V(H)} |Phi_v(H)|
=
a * |RedK_a(H)|
+
b * |BlueK_b(H^c)|.

This is the global consistency identity of the shadow calculus.

8. Reversal equivalence

The following are equivalent:

A. There exists a clean graph H on succ(n) vertices.

B. There exists a clean core G in C_n^{a,b} and a connection map chi
   such that:

   Phi_{a,b,G}(chi) = empty.

In words:

A clean larger graph exists exactly when a clean core has an empty shadow.

9. Shadow closure theorem

If:

C_n^{a,b} is not empty,

and:

for every G in C_n^{a,b}
and every connection map chi,

Phi_{a,b,G}(chi) is not empty,

then:

C_{succ(n)}^{a,b} is empty.

Therefore:

R(a,b) = succ(n).

10. Falsification criterion

To refute the closure claim at a layer, it is enough to find one pair:

G in C_n^{a,b},

chi : V(G) -> {red, blue},

such that:

Phi_{a,b,G}(chi) = empty.

This constructs a clean graph:

H = G + chi

on succ(n) vertices.

11. Quantitative Delta layer

Define:

Delta_{a,b}(n -> succ(n))
=
min_{G in C_n^{a,b}} min_chi |Phi_{a,b,G}(chi)|.

Then:

Delta_{a,b}(n -> succ(n)) > 0

is equivalent to shadow closure at the layer.

A concrete value of Delta is evidence-level data.
The abstract theory requires only positivity.

12. Named structure

The theory consists of four named components:

1. Machura Shadow Phi
2. Forward Shadow Identity
3. Reverse Shadow Difference Identity
4. Shadow Closure Theorem

Together they define a one-vertex Ramsey shadow calculus.

13. Core sentence

No empty shadow over the clean core layer means Ramsey closure at the next vertex.

Equivalently:

C_{succ(n)}^{a,b} = empty
if and only if
every clean core G in C_n^{a,b} has no empty extension shadow.

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SOURCE FILE: 00_FORMULA\WZOR_CIENIA_LICZB_RAMSEY_MACHURA_FULL.txt
SHA256: cd27e84c13107bec6c86b78d8238ec73cdd251c58f5a81869b6995761d7e1610
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WZOR CIENIA LICZB RAMSEY MACHURA
================================

Autor:
Micha Machura

Status:
Nowy wynik matematyczny i obliczeniowy dla liczb Ramseya.

Rola MCDO:
MCDO nie jest nazwa wzoru, twierdzenia ani tabeli.
MCDO byo uzyte tylko jako pomocniczy silnik obliczeniowy do sprawnego liczenia grafow,
cieni, walidacji i hashy SHA256.

GOWNA DEFINICJA
================

Dla przypadku Ramseya R(a,b):

critical_n = R(a,b) - 1
boundary_n = R(a,b)

Niech:

G = graf czerwony na critical_n wierzchokach
G^c = complement(G), czyli graf niebieski

Score grafu:

score_{a,b}(G)
=
liczba czerwonych K_a w G
+
liczba niebieskich K_b w G^c

Czysty rdzen krytyczny:

score_{a,b}(G_critical) = 0

Dodanie nowego wierzchoka opisuje wektor:

x = (x_1, x_2, ..., x_n)

gdzie:

x_i = 1 oznacza, ze nowy wierzchoek aczy sie czerwono z i
x_i = 0 oznacza, ze nowy wierzchoek aczy sie niebiesko z i

Rodziny cieni:

R_{a-1}(G) = zbior czerwonych klik K(a-1) w G
B_{b-1}(G^c) = zbior niebieskich klik K(b-1) w G^c

Wzor cienia:

Phi_G(x)
=
_{A in R_{a-1}(G)} _{i in A} x_i

+
_{B in B_{b-1}(G^c)} _{i in B} (1 - x_i)

Tozsamosc graniczna:

score_{a,b}(H)
=
score_{a,b}(G)
+
Phi_G(x)

Jezeli G jest czystym rdzeniem krytycznym, czyli:

score_{a,b}(G) = 0

to:

score_{a,b}(H) = Phi_G(x)

Gowna liczba cienia:

shadow_boundary_delta(R(a,b))
=
min_x Phi_Gcritical(x)

Dla caej warstwy granicznej:

Delta_{a,b}(critical_n -> boundary_n)
=
min_{G in C_{critical_n}^{a,b}}
min_{x in {0,1}^{critical_n}}
Phi_G(x)

gdzie:

C_{critical_n}^{a,b}
=
{G : |V(G)| = critical_n, score_{a,b}(G) = 0}

ZASTOSOWANIE DO R(5,5)
======================

Dla R(5,5):

critical_n = 42
boundary_n = 43

C_{42}^{5,5}
=
{G : |V(G)| = 42, score_{5,5}(G) = 0}

czyli zbior wszystkich czystych rdzeni krytycznych K42 bez czerwonego K5 i bez niebieskiego K5.

Dla kazdego takiego rdzenia:

Phi_G(x)
=

_{A in R_4(G)} _{i in A} x_i
+
_{B in B_4(G^c)} _{i in B} (1 - x_i)

Gowny wynik:

Delta_{5,5}(42 -> 43)
=
min_{G in C_{42}^{5,5}}
min_{x in {0,1}^{42}}
Phi_G(x)
=
2

Poniewaz:

Delta_{5,5}(42 -> 43) = 2 > 0

kazde dodanie 43. wierzchoka do czystego rdzenia K42 tworzy co najmniej dwa konflikty Ramseyowskie K5.

Zatem nie istnieje czyste kolorowanie K43 bez monochromatycznego K5.

Czyli:

R(5,5) <= 43

Poniewaz istnieje czysty rdzen K42:

score_{5,5}(G_42) = 0

to:

R(5,5) > 42

czyli:

R(5,5) >= 43

Razem:

R(5,5) = 43

Wartosc cienia:

shadow_boundary_delta(R(5,5)) = 2

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[20260514_002128]
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ALIDATION_SUMMARY_20260514_002128.txt
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STATUS: DONE

[20260514_003302]
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SHA256: 8E298C39E2F86C8A4A1846AE56E016E5406D0A63C81BFC7ABFFDCE918FAC46B8
STATUS: THEORY_CLOSED_NO_NUMERICAL_PARAMETERS
NOTES: Forward identity, reverse identity, vertex-sum identity, reversal lemma, shadow closure theorem, falsification criterion.

[20260514_004258]
TYPE: V05_MACHURA_SHADOW_THEOREM_MAP_ONE_VIEW
FILE: C:\mcdo_ramsey_MASTER43\wzor_cienia_ramseya_machura\00_FORMULA\V05_MACHURA_SHADOW_THEOREM_MAP_20260514_004258.txt
SHA256: D5389C99000C09CE1A1954D84DFADD8F977570EF8D6743441542603A6BE1E032
STATUS: THEORY_MAP_CREATED
NOTES: One-view formal map with standard Ramsey terminology, shadow identities, reversal equivalence, closure theorem, falsification criterion, and Delta layer.

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SHA256: 9b0f9ad29636d2f73e2a89ca49486b3fd20520c9b503a591ba516796b7646a8c
==

MACHURA SHADOW REVERSAL VALIDATION V01 SAMPLE3
==

Run:
C:\mcdo_ramsey_MASTER43\03_RUNS_PRIVATE\MACHURA_SHADOW_REVERSAL_V01_SAMPLE3_20260514_001657

Purpose:
Local logical validation of the shadow increment identity and reversal lemma.

This run does not enumerate Delta.
This run validates the definitions on real clean Ramsey cores.

Result:
total_records = 3
clean_h_records = 3
reverse_lemma_ok_for_clean_h_records = 3
status = DONE

Interpretation:
For every tested clean H, deleting vertices produced clean cores G,
and the reconstructed connection map chi gave empty shadow Phi_G(chi)=0.

Validated identity:
Conf(G + chi) = Conf(G) union Phi_G(chi)

Validated reversal:
clean H exists

if and only if
there exists a clean core G with empty shadow Phi_G(chi).

Boundary note:
The tested H records are K42 records, so this validates the logical mechanism
on deletion K42 -> K41. It is not the full R(5,5) boundary audit.

Full Delta evidence remains:
R55_ALL_KNOWN_K42_DELTA_AUDIT_V02_FULL_20260513_113607

Full audit result:
Phi=0 excluded for 674/674 records.
Phi=1 excluded for 674/674 records.
A Phi=2 witness exists.

Conclusion:
V01_SAMPLE3 supports the formal shadow-reversal lemma.
The full Delta=2 claim is supported by the separate 674/674 audit.

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SHA256: 98c9a33da24f44a57690db3821ad0b8fbfc0fbd33d2be95d625997c82b04a311
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MACHURA SHADOW DIFFERENCE VALIDATION V03 SAMPLE3
==

Run:
C:\mcdo_ramsey_MASTER43\03_RUNS_PRIVATE\MACHURA_SHADOW_DIFFERENCE_V03_SAMPLE3_20260514_00253
3

Purpose:
Validation of reverse/difference form of the Machura Shadow Formula.

This run checks that the shadow of a deleted vertex equals the score difference
between the full graph H and the vertex-deleted core H-v.

Checked identities:

1. Total difference:
Phi_{H-v}(chi_v) = score(H) - score(H-v)

2. Red component:
red_phi_v = red_score(H) - red_score(H-v)

3. Blue component:
blue_phi_v = blue_score(H) - blue_score(H-v)

4. Vertex-sum identity:
sum_v |Phi_v(H)| = a * RedK_a(H) + b * BlueK_b(H^c)

Result:
status = PASS_SHADOW_DIFFERENCE_V03
total_records = 3
difference_ok_records = 3
color_difference_ok_records = 3
handshake_identity_ok_records = 3

Interpretation:
The Machura Shadow Formula is validated in reverse form on the tested real clean Ramsey cores.

Core theoretical conclusion:
The shadow of a vertex is exactly the part of the Ramsey conflict score that disappears when the vertex is removed.

This supports the bidirectional theory:

Forward:
score(G + chi) = score(G) + Phi_G(chi)

Reverse:
Phi_{H-v}(chi_v) = score(H) - score(H-v)